



Investigation of manganese metal coordination in proteins: a comprehensive PDB analysis and quantum mechanical study

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Abstract

Manganese (Mn) is an important metal that is crucial in biological cell mechanism and function. However, its binding mechanism is poorly characterized. In the present study, we have carried out a detailed statistical analysis of the Mn-containing proteins through analysis of the metal coordination spheres of the vast number of protein crystal structures present in Protein Data Bank. These results reveal that Mn metal predominantly acquires the coordination number of **six and five**. In these predominant six and five coordination spheres, Mn metal is majorly stabilized with **octahedral and square pyramidal** geometries respectively. The **water molecules, aspartic acid, and glutamic acid residues** bonded frequently with Mn metal ions. These results provided useful insights to characterize the very important Mn-containing subset of the proteome. Quantum mechanical results showed that the complexes with coordination number six are predominantly having high interaction energy, which is in good agreement with statistical analysis.

Keywords Manganese · PDB · Quantum mechanical study · Coordination sphere · DFT

Introduction

More than 30% of the proteins contain metals as cofactors to perform their functions. The role of metal ions is crucial for the protein crystal formation as metal ions intervene contact between protein residues. Moreover, the d-block metal cations such as Mn, Fe, Co, Ni, Cu, and Zn are crucial for structural stability and proper cell function [1]. The metals in the proteins contribute to the stability of the protein structure and also play a crucial role in the activity. Particularly, the function of metalloproteins majorly depends on the presence of a specific metal ion [2]. The hard metals named Mg, Ca, and Mn prefer to bind directly with the oxygen-containing amino acids such as Asp, Glu, Asn, and Gln and backbone carbonyl groups [3–6]. Frequently, water molecules also participate in metal coordination, and it stabilizes the metals in a particular

conformation [5, 6]. The explicit properties of each metal, such as electron configuration and size, define the atom composition and geometry of metal-binding sites. Further, these properties also influence the selectivity in the metal coordination sphere and biological function of proteins [7]. Metals participate widely in the variety of metal coordination spheres in proteins. However, these coordination spheres are not specific to particular metal [8]. Delineating the metal coordination in these proteins is one of the utmost importance for understanding the protein functionality and stability [9–11].

In the previous studies, researchers have investigated that protein structures from RCSB Protein Data Bank (PDB) for metalloproteins before 2009. In these studies, they have analyzed the binding properties of different metals, precisely their coordination tendency to bind specific amino acid residues and their preferred coordination numbers [7, 12–17]. Most of these studies focused on the identification of the most common atoms and amino acid residues in the metal geometries. Along with these studies, there are a couple of databases available about metal coordination geometries [18, 19, 20, 21]. Zheng et al. in a study reveals that histidine is the most preferred amino acid in the binding sites, while glutamic acid has the lowest preference [16]. Based on the work of Brylinski et al., aspartic acid has a much higher preference to bind Mn^{2+} . Most of

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the previous studies focused on the preference of amino acid binding and coordination number [22]. However, the preferred geometry and their structural details were studied in scarce. Thus, there is a need for a systematic study of binding site interactions and geometries of Mn-containing proteins.

To best of our knowledge, there are very few studies on Mn coordination geometries. However, there is no systematic study on Mn geometrical parameters, coordination sphere, and the study on stability of coordination sphere of Mn in proteins. In the present study, we have carried out the analysis of the Mn coordination environments on proteins available to the present date. We have focused on the most preferred coordination number and the most common combination of ligands at the metal coordination sphere. The most favored Mn-binding amino acid residues and atoms and also the number of Mn atoms present in proteins for the total data set of Mn-containing proteins are considered in this study. Besides, the quantum mechanical (QM) calculations are performed on the combinations of ligands present in Mn-containing biological systems. The interaction energies of the corresponding Mn coordination sphere are evaluated.

Computational details

PDB data set and analysis

The Protein Data Bank (PDB) was searched for proteins containing at least one Mn atom. The analysis of Mn coordination spheres was carried out on a screened set of proteins deposited in the PDB until March 2019. To make data set scientifically significant, we have removed redundancy in the data set by employing the criterion as protein structures sequence identity to $\leq 95\%$. The final data set containing 1145 proteins is obtained from the downloaded proteins of PDB. To search the large data set, we have written a script in R (R: a language and environment for statistical computing) package to locate the geometrical features of the Mn coordination sphere. The present script scans the given protein structure from C-terminal to N-terminal for the identification of Mn metals. Further, this program also searches for the residues and atoms present within the distance of 3 Å from the Mn metal. All the interactions following these criteria were considered. The raw data obtained was refined and used for further analysis. The properties such as identification of Mn-coordinating residues and Mn-containing atoms, the number of Mn atoms present in each protein, the coordination number of each Mn coordination sphere, and the preferred combinations of ligands at each coordinating sphere were analyzed to understand the deeper insights of Mn metal properties in proteins.

Quantum mechanical study

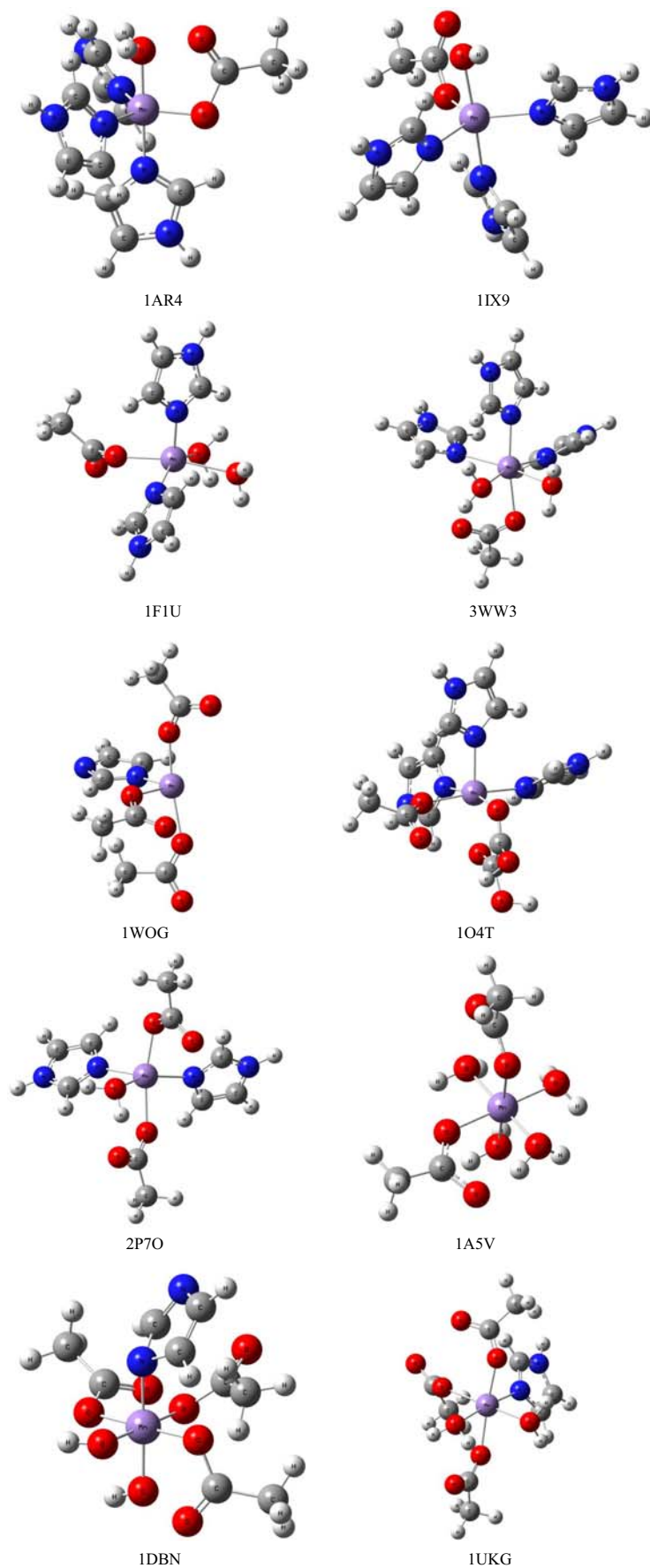
To understand the thermodynamic stability of the Mn-coordinated complexes, we have investigated interaction energies of the most common ligand combinations for various coordination numbers such as four, five and six using sophisticated QM calculations. For QM calculations, model systems were prepared using particular PDB structures for different combination of ligands at the metal coordination site. These models were prepared with Mn metal and the first coordination sphere residues. The amino acid model systems were used as part of conventional modeling of amino acid side chains, such as aspartate (or glutamate), histidine, asparagines, and oxalic acid, were modeled by acetate, imidazole, carboxamide and oxalic acid respectively (Fig. 1). All initial calculations were performed using Gaussian 09 software package [23]. Each of these structures was initially optimized using B3LYP/SDD method. Further, the vibrational frequency analysis calculations were performed with the same method and basis set as those for the geometry optimizations. For transition metal containing complexes, DFT calculations give accurate results [24]. In the present study, we have used SDD basis set that use small core quasi-relativistic Stuttgart/Dresden electron core potentials for transition metals [25, 26, 27]. The Gibbs free energies were calculated at 298.15 K. To obtain the solvation free energies, we performed single-point calculations within the universal solvation model SMD, [28, 29] by using the SMD-B3LYP/SDD//B3LYP/SDD level of theory on the gas phase B3LYP/SDD optimized geometries. Previous studies on protein active sites have shown that dielectric constant (ϵ) of 4 showed good agreements with the experimental results [30, 31, 32]. Thus, in this study, we have also used ϵ as 4 for all solvation model calculations. Free energies of reactant species in solution G_{298}^S were obtained by adding the SMD free energies of solvation to the gas-phase free energies of the residues present in the complex.

Results and discussion

Number of Mn atoms present in proteins

Many proteins are having more than one similar metal atom in their structures. The proteins with mono- and polynuclear metal atoms have their importance in protein structure and function. Herein we have analysed the number of Mn atoms present in the proteins. In the present data set, many proteins contain more than one Mn metal. Figure 2 depicts the number of Mn metals present in proteins. A total of 62% of proteins in the present data set are mononuclear Mn proteins. Around 28% of the proteins are binuclear Mn-containing proteins. The polynuclear such as tri- and tetranuclear Mn proteins are also common in the

Fig. 1 Various models studied with Mn metal and the first coordination sphere residues using QM calculations of four, five, and six coordination numbers. The amino acid model systems were used as part of the conventional modeling of the amino acid side chains, such as aspartate (or glutamate), histidine, asparagine, and oxalic acid, and were modeled by acetate, imidazole, carboxamide, and oxalic acid, respectively



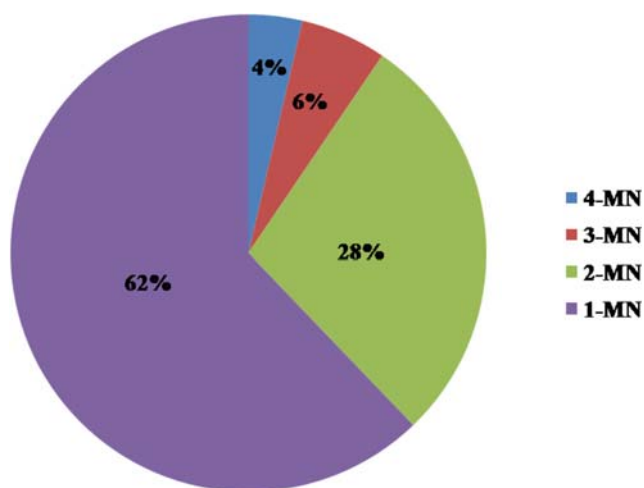


Fig. 2 The number of Mn metals present in each protein in the present data set of proteins

PDB with 6% and 4% occurrence respectively. In polynuclear Mn-containing proteins, the main structural feature Mn-Mn bond is observed in most of the proteins. This analysis shows that proteins with mononuclear Mn are more preferable in comparison to polynuclear coordination spheres.

The coordination numbers of Mn-containing proteins

The most important property to analyze in metal proteins is the coordination number of the metal. In the present study, we have analyzed all possible coordination numbers of Mn-containing proteins. Further, we have also analyzed the coordination number and preferable geometry of the Mn-containing proteins. The results of the coordination number tendencies of all examined Mn metals are summarized in Fig. 3. The Mn metal possesses various coordination numbers with residues and water molecules in the protein structures.

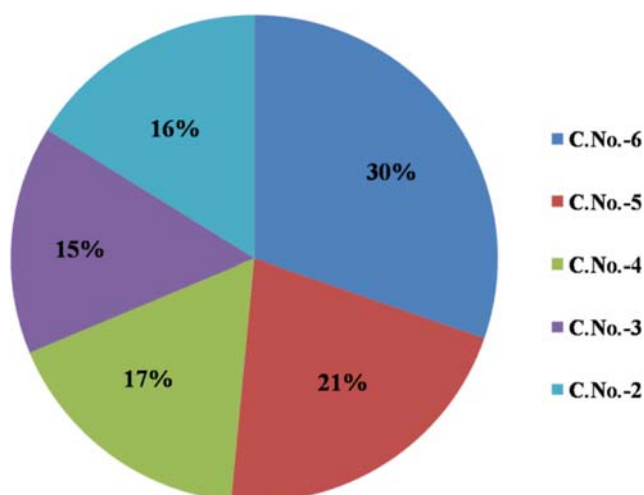


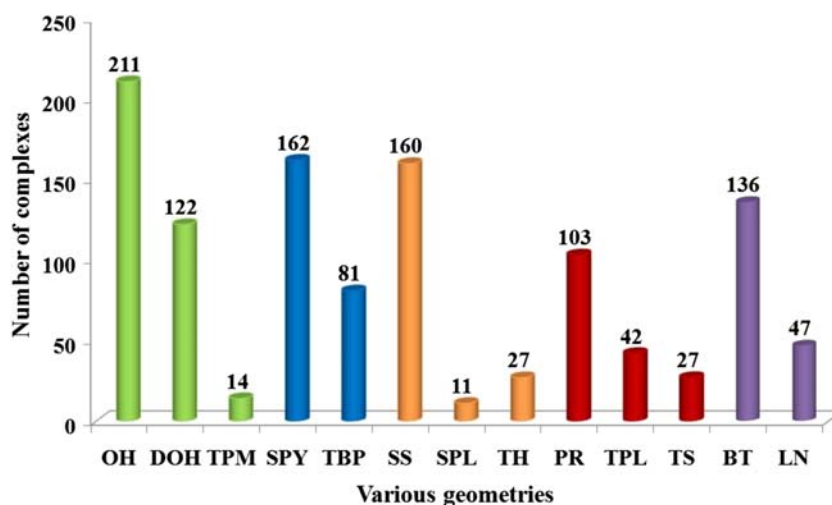
Fig. 3 The results of the coordination number tendencies of all examined Mn metals complexes considered in the present data set of proteins

Mn metals have coordination numbers from six to two in the proteins. This result reveals that Mn metal predominantly acquires coordination number of six and five. In particular, around 30% of Mn metals in the considered data set are preferred six coordination spheres. In addition to coordination number six complexes, around 21% of Mn metals are bonded with five residues with coordination number five. The coordination number of four, three, and two is also available in the proteins with 17%, 15%, and 16%, respectively. The above analysis reveals that the coordination number of six and five is preferable for the Mn-containing proteins to attain stability in the proteins. It is also clear that six coordination geometries are preferable for efficient biological function of Mn-containing proteins.

There are a variety of geometries that are possible for each coordination number in general. Thus, it is essential to understand the most favorable geometry. As we discussed above, Mn metals showed coordination numbers six to two. We have also analyzed the various geometries of Mn metal acquired in the various coordination spheres. Figure 4 depicts the number of complexes showed for a particular geometry of Mn metal complexes in the proteins. In this analysis, we have considered possible geometries for coordination numbers shown in Fig. 3.

To get atomistic details, we have also analyzed the possible geometries in each coordination number. For the most preferred coordination number six, we have considered possible octahedral (OH), distorted octahedral (DOH), and trigonal prism geometries. In these data sets, most of the complexes form octahedral geometry with 211 structures, while 122 structures adopted distorted octahedral geometry. For the preference of OH geometry over DOH geometry, the reason may be the planarity of OH geometry which will provide more stability to the Mn coordination sphere than Mn in the non-planar geometry. In the coordination number, five square pyramidal and trigonal bipyramidal geometries are considered. In these square pyramidal geometry is favored with 162 structures than trigonal bipyramidal. Interestingly, in coordination number four, seesaw geometry (160 structures) is more favorable over the regular tetrahedron (27) and square planar (11) geometries. In coordination number three, pyramidal, trigonal planar, and T-shape geometries are taken into account. Herein this, Mn metal complexes acquired more pyramidal geometries with 103 structures, while T-shape and trigonal planar geometries have appeared with 27 and 42 number of structures respectively. The bent and linear geometries are considered for coordination number two. Mn complexes have more a number (136) of bent structures than structures of linear geometry (47) in coordination number two. Finally, this analysis clearly shows that octahedral geometry is more predominant with 211 structures for the Mn metal containing geometries in the proteins. Further, square pyramidal geometry in coordination number five and seesaw geometry in coordination number four are predominant with 162 and 160 structures

Fig. 4 The number of complexes found for each geometry of various coordination numbers present in Mn metal complexes considered in the present data set of proteins. Herein octahedral (OH), distorted octahedral (DOH), trigonal prism (TPM), square pyramidal (SPY), trigonal bipyramidal (TBP), seesaw (SS), square planar (SPL), tetrahedral (TH), pyramidal (PR), trigonal planar (TPL), T-shape (TS), bent (BT), and linear (LN)



respectively in this study. Besides, the pyramidal and bent structures are the most frequently occurred structures for coordination numbers three and two respectively.

The nature of Mn coordination spheres

Apart from coordination number and geometry, the nature of atoms and residues present in the coordination sphere is also interesting. So we have carried out the analysis of the nature of Mn coordination spheres.

Figure 5 depicts the percentage of Mn-binding site atoms and various ligands in the Mn protein structures present in the PDB. These results (Fig. 5a) clearly show that oxygen and nitrogen atoms are bound with Mn protein complexes in most of the protein structures of Mn complexes present in PDB. Precisely, oxygen is the most common Mn-containing atom (81%); this result arises the presence of oxygen in Mn-bound water molecules and Asp and Glu amino acid residues and is also present in some protein inhibitor molecules

and ligands. In addition to oxygen, nitrogen accounts for 18% of the Mn-coordinated atoms. Nitrogen presents mainly in His amino acid and other ligand molecules coordinated to Mn metals in proteins. Apart from O and N coordinated to Mn metal other atoms are having very little percentage (1%) and are almost negligible.

Figure 5b shows the percentage of Mn-binding ligands in the Mn-containing proteins present in PDB. The result reveals that water molecules are the most common residue in Mn coordination sites, which occupies 26% of all Mn-containing protein coordination sites. Asp is the second most common ligand coordinated with Mn metal in proteins that account for 24% of all the Mn-ligand coordination sites. In Mn-ligand coordination spheres, His is the third most common amino acid residue account for 16%. Glu and ligands/inhibitor molecules occupy 13% and 12% in the total Mn-ligand coordination spheres respectively. Other amino acids and residues, particularly with inhibitors and non-peptide molecules, occupy a very less percentage of Mn-ligand

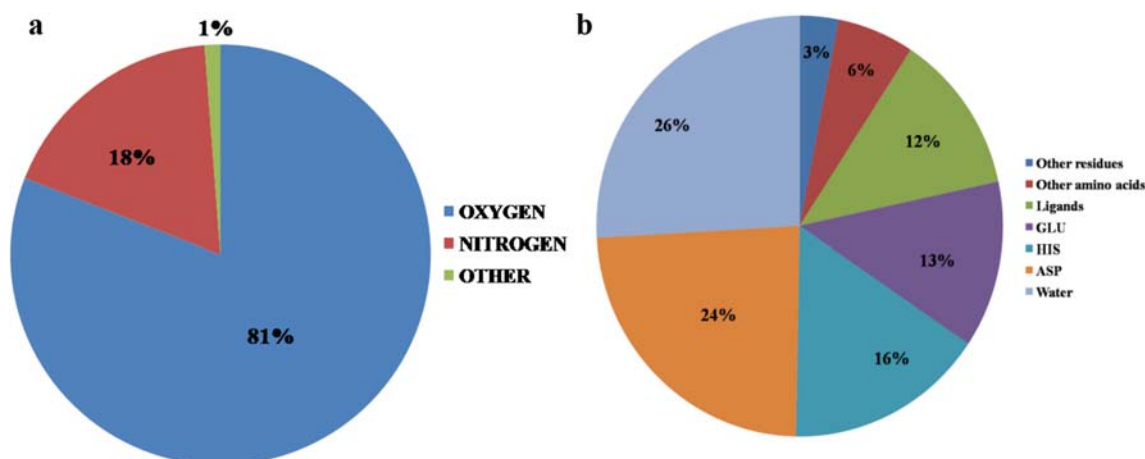


Fig. 5 The percentage of Mn-binding site atoms (a), amino acid residues and various ligands (b) present in Mn metal complexes considered in the present data set of proteins

coordination spheres such as 6% and 3% respectively. The present results are in rough agreement with some of the previous studies on Mn proteins with different older data sets [33, 34, 35]. Some of the studies showed that His and Cys coordination with Mn is more significant. In this study, we have also analyzed the importance of water molecules in Mn protein complex interactions. The presence of water molecules is important of catalytically active Mn metal coordination spheres in biological systems. The H₂O molecules can also act as ligand and as a critical component in a variety of biological relevant enzyme-catalyzed reactions.

Quantum mechanical results

In support of statistical analysis of Mn-coordinated complexes, we have also performed QM calculations on most common Mn complexes for the coordination number of four, five, and six. This study evaluates the thermodynamic stability of Mn-coordinated complexes. We aimed to correlate the relation between interaction energies and selected complexes coordination spheres and relative number of proteins appeared in statistical analysis.

Figure 6 depicts the interaction energies of selected Mn complexes. The interaction energy (IE) calculated as the difference of solvation Gibbs free energies and total energy of the individual residues present in a particular complex. These results show some interesting outcomes. All complexes are shown interaction energies between -5.00 kcal/mol and -18.06 kcal/mol. In this study, six coordinated complexes with DOH, OH geometries found to be more stable with -17.15 kcal/mol to -18.06 kcal/mol interaction energies. Further, complexes with coordination number four having

SPL and SS geometries have interaction energy -11.87 kcal/mol and -9.88 kcal/mol. In coordination number five, SPY geometrical complexes showed quite good interaction energy such as -10.91 kcal/mol and -7.47 kcal/mol. However, TBP complexes are found to be least stable with -4.98 kcal/mol and -5.32 kcal/mol interaction energy in this study. Overall, complexes with coordination number six showed strong stability followed by SPL, SS, and SPY in coordination number four and coordination number five complexes. This is in good agreement with statistical results where OH, SPY, and SS geometries are predominant (Fig. 4). However, in QM calculations, SPL geometry complexes showed significant interaction energy, wherein statistical analysis, those structures are found to be less in number. On the other hand, SPY geometry is the second most predominant in statistical analysis, but in QM results, this complex showed low stability. These deviations between statistical analysis and QM results may be due to the absence of a specific protein environment and second and higher coordination sphere amino acids in the model complexes.

Globally, there is a correlation between the number of occurrences of each coordination sphere, and its interaction energy was found for higher coordination number complexes. This analysis clearly shows that the protein environment and second coordination sphere residues have a significant effect on the stability of coordination geometries of Mn complexes in biological systems.

We have analyzed the structural changes of considered model structures from PDB structures. Broadly, there are three different types of Mn, and bonded atoms are observed such as Mn-O, Mn-Ow, and Mn-N. In Mn-O bond, the oxygen considered from various model residues. In Mn-Ow type bond,

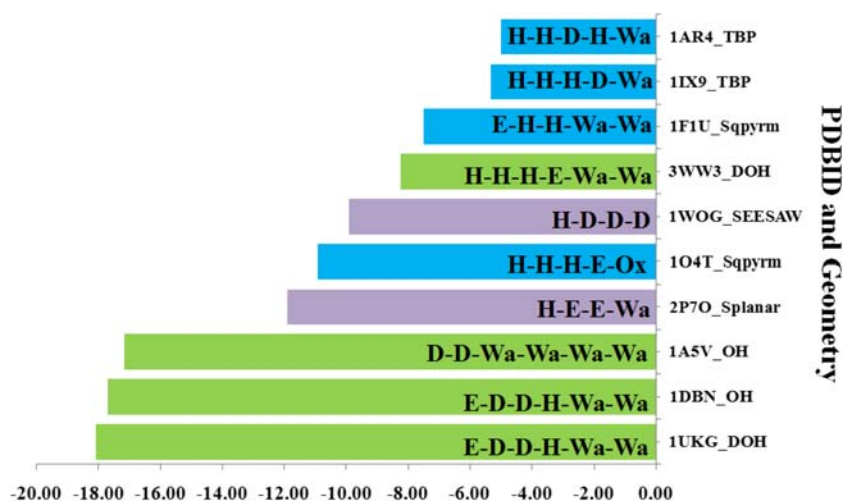


Fig. 6 QM interaction energies calculated as solvation Gibbs free energies (ΔG_{sol} in kcalmol^{-1}) at the SMD-B3LYP/SDD//B3LYP/SDD level of theory for the most common Mn coordination spheres. The letters indicated represent the one-letter code for each amino acid, Wa represents a water molecule, whereas Ox represents oxalic acid residue.

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Herein octahedral (OH), distorted octahedral (DOH), trigonal prism (TPM), square pyramidal (SPY), trigonal bipyramidal (TBP), seesaw (SS), square planar (SPL), tetrahedral (TH), pyramidal (PR), trigonal planar (TPL), T-shape (TS), bent (BT), and linear (LN)

the oxygen bonded with Mn is from the water molecules. After optimization distances of the above mentioned three bonds are slightly changed in comparison to the distances from PDB structures. In this study, Mn-O bond distance in PDB structures varies from 1.83 Å to 2.32 Å. After optimization the bond distance decreased, and those structures fall between 1.79 Å and 1.98 Å. On the other hand, Mn-Ow bond distance also showed similar behavior as Mn-O bond distance; these bonds are in PDB between 1.81 Å and 2.25 Å; after optimization they decreased and appeared between 1.79 Å and 2.09 Å. The Mn-N bond distance in PDB is between 1.87 Å to 2.25 Å, and after optimization the bond distance is decreased and resulted between 1.95 Å to 2.12 Å.

In average from the ten structures considered in this study, the Mn-O bond distance is changed from 2.06 Å to 1.89 Å after optimization. Further, average Mn-Ow bond distance is changed from 2.03 Å to 1.97 Å, and Mn-N bond distance is from 2.03 Å to 1.98 Å. These results reveal that the change in the bond distance between Mn and its bonded atoms is very less (< 0.1 Å). Thus the model systems considered in this study are reliable with Mn coordination geometries in the PDB.

Further, we have also analyzed the relation between the different types of proteins and coordination numbers of Mn-containing coordination spheres. In this study, we have observed mainly six different classes of proteins, namely, hydrolase, isomerase, ligase, lyase, transferase, and oxoreductase. In this analysis, hydrolase proteins showed C. No. 6 and 5 structures around 60%; similarly isomerase type of proteins also have more with C. No. 6 (~ 50%). The ligase and lyase family proteins have Mn-containing coordination spheres with C. No. 6 of 48% and 39% respectively. The structures with C. No. 5 are around 27% in lyase proteins. On the other hand, transferase proteins have C. No. 6 and C. No. 5 as 50% and 20% respectively. In comparison to the above protein types, oxidoreductase proteins have 42% of structures with C. No. 5, and 33% of structures showed C. No. 6.

Overall, all types of proteins showed C. No. 6 as the more preferred coordination followed by C. No. 5. The structure with coordination numbers four and three are very less in all proteins considered in this analysis. These observations are also good in correlation with PDB results. We can also observe that there is no clear correlation between types of proteins and coordination number of Mn-containing protein structures.

Conclusion

In the present study, we have carried out a detailed statistical analysis of metal coordinated spheres of Mn-containing proteins present in PDB. The statistical results reveal that Mn metal predominantly acquires coordination number of six and five. The Mn metal found to be more in octahedral and square pyramidal geometries in six and five coordination

respectively. Mononuclear Mn is more preferable in the proteins in comparison to polynuclear coordination spheres. The oxygen and nitrogen atoms bonding to Mn atom that is more in number may be due to the hard nature of Mn metal. On the other hand, Mn bound frequently with aspartic acid, glutamine, and water residues. These results provide useful insights into this difficult to characterize but very important Mn-containing subset of the proteome. QM results showed that complexes with coordination number six are predominant with high interaction energy, which is in good agreement with statistical analysis.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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